Examiner only

7. (a) In dry conditions, a Formula One (F1) car can accelerate from 0 to 30 m/s in 1.5 seconds in a straight line.

(i) State the change in velocity. [1]

Change in velocity = ... m/s

(ii) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration of the racing car. [2]

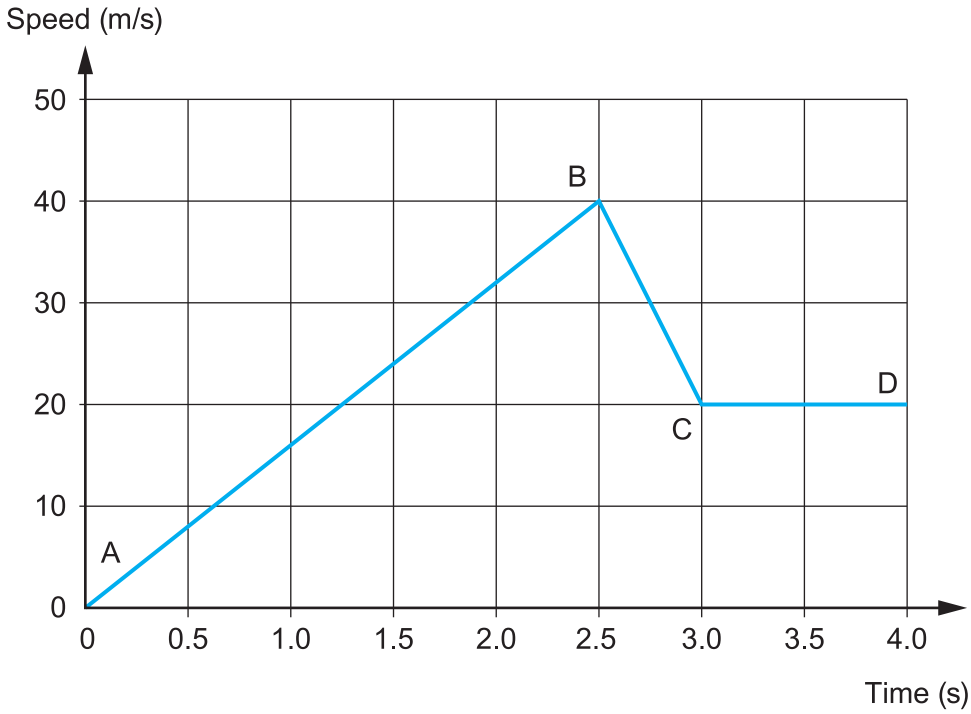
Acceleration = ... m/s^2

(b) The photograph shows F1 cars lined up on the grid.

Sep Year 11 FT 2022_Q7

Examiner only

The graph below shows how the speed of a F1 racing car changes at the start of a race as it leaves the grid and goes around the first bend.



(i) **Use data from the graph** to describe the motion of the car during the time shown. No calculations are required. [6 QER]

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(ii) The car travels 85 m during the time shown by the graph.
Use an equation from page 2 to calculate the mean speed of the car during this time. [3]

Mean speed = ... m/s

12